Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

Course Title: Basic Electrical and Electronic Circuit Lab

Course Code: EEE 0713 1102W

Experiment 06

Verification of Maximum Power Transfer Theorem for a DC Network

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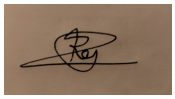
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Verification of Maximum Power Transfer Theorem for a DC Network"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit Lab (EEE 0713 1102W)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

Sylhet, Bangladesh

Date: August 31, 2026



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Objective

- To understand the Maximum Power Transfer Theorem and determine the load resistance at which a DC source network delivers maximum power to a variable load.
- To experimentally plot load power against load resistance and locate the point of maximum power transfer.
- To compare the measured results against theoretical calculations and evaluate the percentage error.

Introduction

When a DC source network – with some fixed internal (Thevenin) resistance – is connected to a variable load, a natural question is what load resistance pulls the most power out of it. The **Maximum Power Transfer Theorem** answers this question directly: the load draws maximum power when its resistance is made equal to the source's own internal (Thevenin) resistance. This experiment builds a Thevenin-equivalent source circuit, sweeps a variable load resistance across a range of values, measures the load current and power at each setting, and locates the load resistance at which the power actually peaks.

Theory

Statement of the Theorem

A linear, two-terminal DC network delivers maximum power to a variable external load resistance R_L if and only if R_L is made equal to the Thevenin equivalent resistance R_{th} of the network as seen from the load terminals:

$$R_L = R_{th}.$$

Derivation

If we replace the source network with its Thevenin equivalent (V_{th} in series with R_{th}) driving a load R_L, the load current works out to

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad P_L = I_L^2 R_L = \frac{V_{th}^2 R_L}{(R_{th} + R_L)^2}.$$

To find the R_L that maximizes P_L, we differentiate with respect to R_L and set $\dfrac{dP_L}{dR_L} = 0$, which leads to the condition $R_L = R_{th}$. At that point,

$$I_{L(\max)} = \frac{V_{th}}{2R_{th}}, \qquad P_{L(\max)} = \frac{V_{th}^2}{4R_{th}}.$$

It's worth noting that at this matched condition, only half of the power the source generates actually makes it to the load – the other half is lost inside the source's own internal resistance R_{th}. So the **efficiency is only** 50% right at maximum power transfer, which shows that getting the most power out of a source and getting the most efficiency out of it are two different goals that can't both be met at once.

Table 1: List of Equipment and Components Used

Item Description	**Specification / Rating**	**Quantity**
DC Power Supply (Thevenin source, V_{th})	8 V	1
Fixed Series Resistor R_i (= R_{th})	663 Ω	1
Decade Resistance Box (Variable Load R_L)	0 Ω–5 kΩ	1
Digital Multimeter (DMM)	DC Volts / DC Amps	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

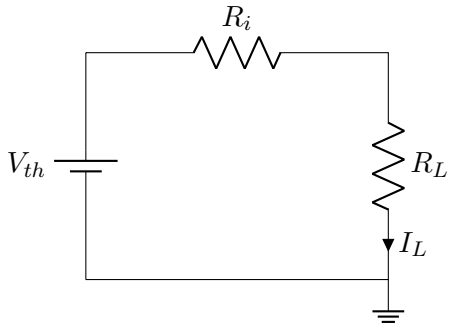


Figure 1: Thevenin-equivalent source ($V_{th} = 8\,\text{V}$, $R_i = 663\,\Omega$) driving a variable decade-box load R_L, used to trace the load-power curve.

Apparatus and Components

Circuit Diagram

Experimental Procedure

1. Build the Thevenin-equivalent source circuit shown in Figure 1, using $V_{th} = 8\,\text{V}$ and an internal series resistance $R_i = 663\,\Omega$.
2. Connect the decade resistance box across the output terminals so it acts as the variable load R_L.
3. Starting from a low value, step R_L up through a range of settings that go well below R_i and well above it.
4. At each setting, measure the load current I_L and work out (or measure directly) the corresponding load power P_L.
5. Plot P_L against R_L, find the load resistance where the power peaks, and check it against the theoretical prediction $R_L = R_i$.

Theoretical Calculations

With $V_{th} = 8\,\text{V}$ and $R_i = R_{th} = 663\,\Omega$, the theoretical load current and power at any load setting follow

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad P_L = I_L^2 R_L,$$

and the theoretical maximum power occurs exactly at $R_L = R_{th} = 663\,\Omega$:

$$P_{L(\max)} = \frac{V_{th}^2}{4R_{th}} = \frac{8^2}{4 \times 663} = \frac{64}{2652} = 24.13\,\text{mW} \approx 24.12\,\text{mW}.$$

These formulas were evaluated at each of the seven load-resistance settings used experimentally, and the results are reported alongside the measured values in Table 2.

Data Collection

Table 2: Experimental Measurement Log for Maximum Power Transfer Evaluation ($V_{th} = 8\,\text{V}$, $R_i = 663\,\Omega$)

S.No.	R_L (Ω)	Measured I_L (mA)	Calculated I_L (mA)	Observed P_L (mW)	Calculated P_L (mW)
1	99.6	10.70	10.50	11.30	10.92
2	327	8.22	8.08	22.10	21.80
3	460	7.23	7.12	24.00	23.30
4	**663 (Peak)**	**6.14**	**6.03**	**24.90**	**24.12**
5	973	4.97	4.89	23.90	23.20
6	2152	2.88	2.84	17.80	17.30
7	3232	2.07	2.05	13.90	13.60

Note: Data transcribed from the instructor-reviewed handwritten laboratory data sheet (Group 04). Measured current I_L was read directly with the DMM; observed power P_L was computed from the corresponding measured load voltage and current.

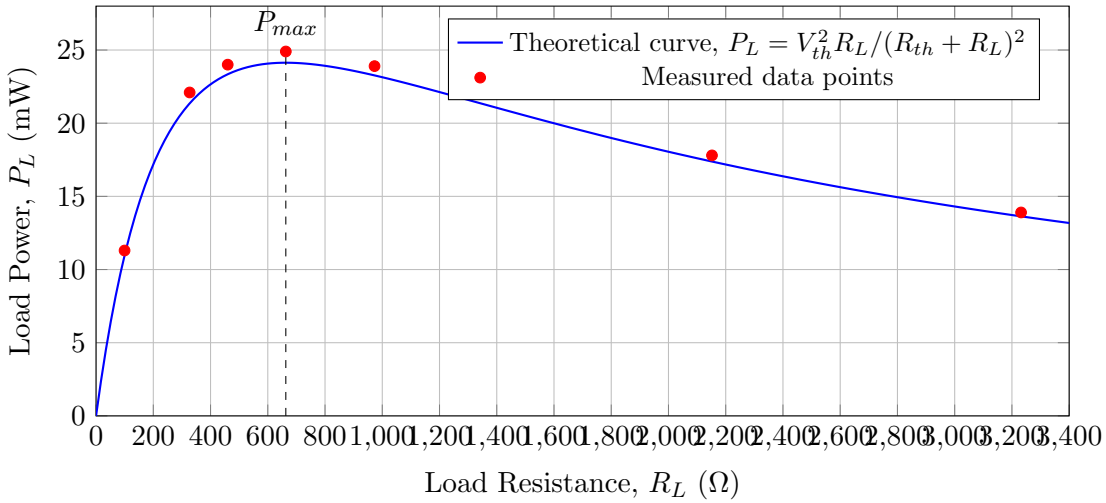


Figure 2: Load power P_L versus load resistance R_L, plotted from the measured data of Table 2. The peak occurs at $R_L \approx 663\,\Omega = R_i$, consistent with the Maximum Power Transfer Theorem.

Result Analysis

Looking at Table 2, the observed load power hits its highest value right at $R_L = 663\,\Omega$ ($P_L = 24.90\,\text{mW}$), which is exactly the internal resistance R_i – just what theory predicts. Comparing this peak to the theoretical maximum,

$$\%\ \text{Error} = \left|\frac{P_{L(\text{max,calc})} - P_{L(\text{max,obs})}}{P_{L(\text{max,calc})}}\right| \times 100\% = \left|\frac{24.12 - 24.90}{24.12}\right| \times 100\% \approx 3.23\%.$$

As Figure 2 shows, the power curve rises on both sides of $R_L = 663\,\Omega$ and there's only one peak, right where it should be, which confirms the Maximum Power Transfer Theorem.

Discussion and Sources of Error

The percentage error at the peak was about 3.23%, so overall the results back up the theory reasonably well. Still, a few things in the real setup keep it from matching the ideal calculations exactly:

1. **Resistor Tolerance:** R_i is not exactly its labeled value – every resistor is only guaranteed to within a manufacturer tolerance – so there's already a small gap between what the theory assumes and what's actually in the circuit.

2. **Power Supply Regulation:** The DC supply output can shift slightly once it's under load, which adds a bit of error at each load setting.

3. **Multimeter Loading:** An ammeter isn't a true zero-resistance short – it has some small burden resistance – and a voltmeter isn't a true infinite resistance either, so just connecting the meters changes the circuit slightly from what's assumed in the calculations.

4. **Decade Box and Contact Resistance:** The decade box can only be set in discrete steps, and the breadboard and jumper wires add a bit of contact resistance that the ideal model doesn't account for. This shows up most clearly near the peak of the power curve, where R_L is close to R_i and the curve is changing shape quickly.

Conclusion

This experiment confirmed the Maximum Power Transfer Theorem for a DC network. The load power measured across seven R_L settings traced out the expected curve, peaking right at $R_L = 663\,\Omega$, which matches the source's internal resistance R_i almost exactly, with a percentage error of only about 3.23% at the peak. That confirms the matching condition $R_L = R_{th}$ predicted by theory, and it also illustrates the trade-off between power and efficiency mentioned earlier – getting the most power out of the source means running at only 50% efficiency, since half of the power is used up inside the source itself.

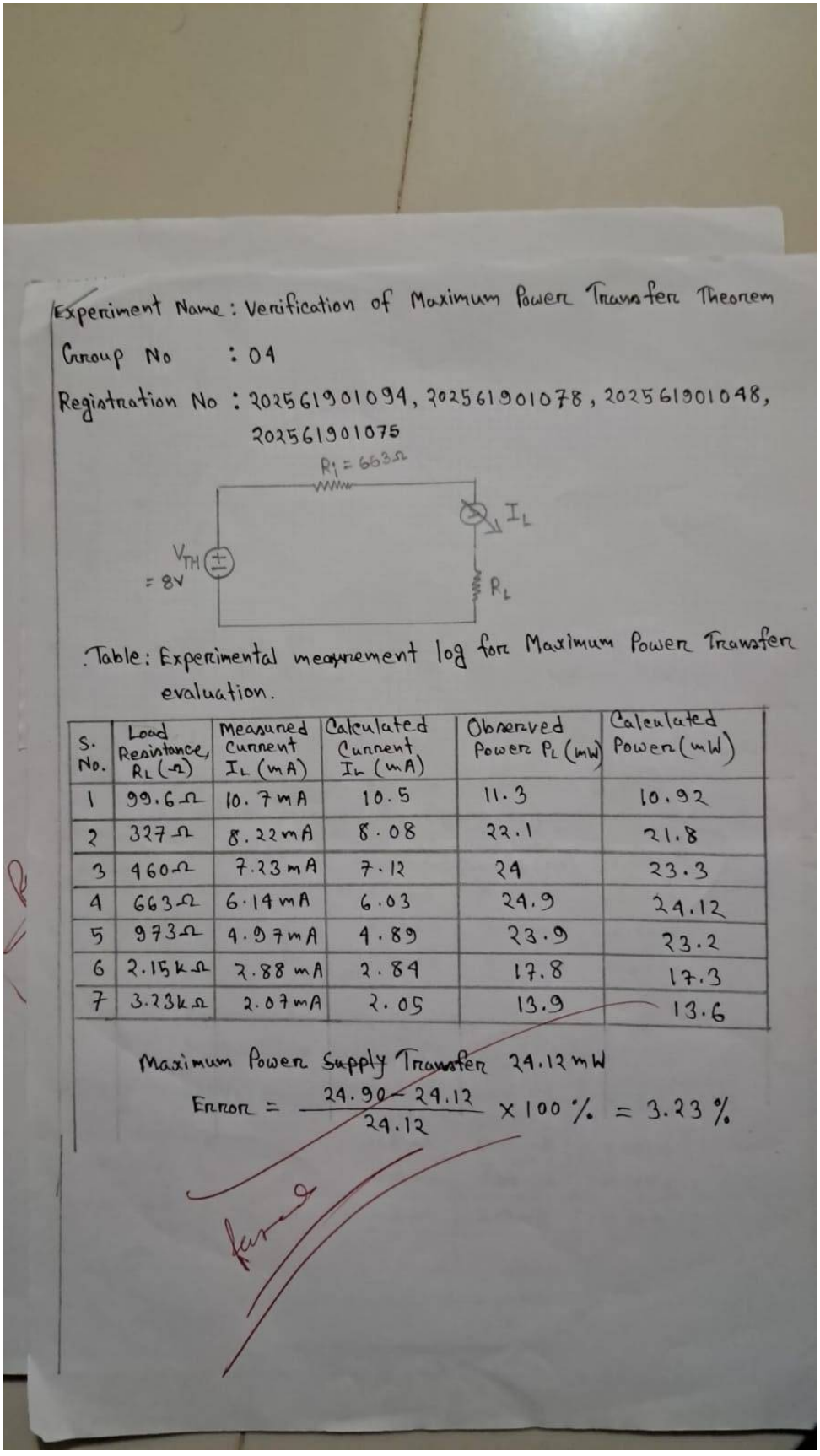
References

1. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education. (Chapter 4: Circuit Theorems).

2. Hayt, W. H., Kemmerly, J. E., & Durbin, S. M. (2012). *Engineering Circuit Analysis* (8th ed.). McGraw-Hill Education. (Chapter 5: Useful Circuit Analysis Techniques).

3. Boylestad, R. L. (2015). *Introductory Circuit Analysis* (13th ed.). Pearson. (Chapter 9: Network Theorems).

Appendix: Signed Data Sheet



Experiment Name : Verification of Maximum Power Transfer Theorem

Group No : 04

Registration No : 202561901094, 202561901078, 202561901048, 202561901075

Table: Experimental measurement log for Maximum Power Transfer evaluation.

S. No.	Load Resistance, R_L (Ω)	Measured Current I_L (mA)	Calculated Current I_L (mA)	Observed Power P_L (mW)	Calculated Power (mW)
1	99.6 Ω	10.7 mA	10.5	11.3	10.92
2	327 Ω	8.22 mA	8.08	22.1	21.8
3	460 Ω	7.23 mA	7.12	24	23.3
4	663 Ω	6.14 mA	6.03	24.9	24.12
5	973 Ω	4.97 mA	4.89	23.9	23.2
6	2.15 k Ω	2.88 mA	2.84	17.8	17.3
7	3.23 k Ω	2.07 mA	2.05	13.9	13.6

Maximum Power Supply Transfer 24.12 mW

$$\text{Error} = \frac{24.90 - 24.12}{24.12} \times 100\,\% = 3.23\,\%$$

Figure 3: Instructor-signed laboratory data sheet for this experiment.